# Part A: Conceptual Foundation (Theory)

**1. What is Regularization in Machine Learning? Why is it needed?**

Regularization is a technique that adds a penalty to a model to control its complexity. It is needed to prevent overfitting and improve the model's performance on unseen data.

**2. Difference between Ridge Regression (L2) and Lasso Regression (L1).**

- Ridge Regression (L2): Adds a penalty based on the squared values of coefficients. It reduces coefficient values but usually does not make them exactly zero.
- Lasso Regression (L1): Adds a penalty based on the absolute values of coefficients. It can make some coefficients exactly zero, helping with feature selection.

**3. What is Cross-Validation and why is it important?**

Cross-validation is a technique used to evaluate a model by dividing data into multiple training and validation sets. It helps estimate model performance more reliably and identify whether the model generalizes well to unseen data.

**4. Explain the following cross-validation techniques.**

a. K-Fold Cross-Validation

The dataset is divided into K equal folds. The model is trained K times, using one fold for validation and the remaining folds for training each time. The final performance is calculated by averaging the results.

b. Stratified K-Fold Cross-Validation

Stratified K-Fold maintains a similar distribution of target groups across all folds. For regression, target values can be divided into bins before applying this technique.

c. Leave-One-Out Cross-Validation (LOOCV)

LOOCV uses one data point for validation and all remaining data points for training. This process is repeated for every data point, making it computationally expensive for large datasets.

d. Time Series Split

Time Series Split divides data according to chronological order. The model trains on past observations and validates on later observations, making it useful for forecasting future values.

**5. Why are tree-based models less sensitive to feature scaling?**

Tree-based models make decisions using feature thresholds rather than distances or coefficient magnitudes. Scaling features usually preserves their relative ordering, so it generally does not significantly affect the model's splits.